

TUTORIAL SHEET

TOPIC :GRAPHS

Question 1

- A. Define the following terms with examples: Degree of a vertex, Path and cycle, Connected and disconnected graphs
- B. Represent the following graph using both adjacency matrix and adjacency list:
Vertices: A, B, C, D
Edges: (A,B), (A,C), (B,C), (C,D)
- C. Perform Breadth-First Search (BFS) and Depth-First Search (DFS) on the following graph starting from vertex A:
Vertices: A, B, C, D, E
Edges: (A,B), (A,C), (B,D), (C,E), (D,E)
Show the order of traversal for both.

Question 2

- A. Compare Prim's and Kruskal's algorithms.
- B. Apply Kruskal's Algorithm to find the Minimum Spanning Tree (MST) for the following weighted graph: Vertices: A, B, C, D, E , Edges: (A,B,4), (A,C,2), (B,C,5), (B,D,10), (C,E,3), (E,D,7)
- C. Consider the following weighted graph: Vertices: A, B, C, D, E, F ,Edges and their weights:(A,B,4), (A,C,2), (B,C,5), (B,D,10), (C,E,3), (E,D,4), (D,F,11)

Using Dijkstra's algorithm, find the shortest path from vertex A to all other vertices. Draw the shortest path tree starting from A. What is the shortest path from A to F and its total weight? Explain briefly how Dijkstra's algorithm determines the next vertex to visit at each step.